Daniel Fausto Franco

Software Developer (Full-Stack · TypeScript) | Firmware · Electronics · Robotics

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Portfolio

PROFESSIONAL SUMMARY

Software Developer with **6+ years of experience** building and supporting solutions at scale (POS, kitchen management, dashboards and autonomous robots) with **React, TypeScript, Angular, Vue.js, Node.js, SQL and Python**, backed by a strong **Biomedical Engineering** foundation: embedded firmware, PCB design and robotics (ROS). A hybrid profile able to deliver products end-to-end —from the **microcontroller to the frontend**— with experience in payment integrations and **AI (OpenAI/Gemini)**. Focused on software quality, clean architecture and continuous improvement.

TECHNICAL SKILLS

Languages: TypeScript · JavaScript (ES6+) · Python · C++ · C · C# · SQL

Frontend: React · Angular (v15–v20) · Vue 3 · Next.js · RxJS · NgRx · Redux Toolkit · HTML5 · CSS3/SASS · TailwindCSS · Bootstrap · Material-UI · Recharts · ag-grid

Backend: Node.js · Express · Flask · REST APIs · MongoDB · SQLite · MySQL · PostgreSQL · Sequelize

Cross-platform: Electron · Cordova · peripherals (printers, scanners, cash drawers) · Socket.IO · SignalR

Applied AI: OpenAI API · Google Gemini · OCR (Tesseract.js) · Gazebo simulation/training

> **Hardware · Firmware · Robotics:** C/C++ firmware · microcontrollers (**PSoC 1/4/5/6, PIC, ATmega, ESP32, Arduino, Raspberry Pi**) · FPGA (Basys) · **I2C, SPI, UART, CAN, BLE** · motor drivers & actuator control · **ROS / RVIZ / Gazebo** · PCB design (Eagle, Proteus, Multisim, OrCAD) · CAD/3D & additive manufacturing (SolidWorks, Inventor) · instrumentation (LabVIEW, MATLAB) · SMD soldering

Testing: Jest · Playwright · Protractor · Cypress · Mocha · Karma/Jasmine

DevOps / Tooling: Git · GitLab · Docker · CI/CD · Webpack · Vite · Jira · Slack

Methodologies: SOLID · Clean Architecture · Agile / Scrum

WORK EXPERIENCE

Software Developer — GA Tech (assigned to Xenial / Global Payments) *Feb 2023 – Present · Remote*

Platform used by global brands: Burger King · Denny's · SeaWorld · Carrefour · H&M · Volkswagen · Royal Caribbean · Ironman

- Development and **support of the Genius POS and XKM (Kitchen Management)** systems by Xenial/Global Payments, used by global chains across restaurants, retail, automotive, cruises and events.
- Stack: **Angular 20 + TypeScript + SQLite** (POS) and **Vue 3 + Electron** (XKM); contributed to the **cashless (payment) integration**.
- Automated testing (**Jest, Playwright**), hotfixes, code reviews and multi-brand production support; Agile ceremonies (Jira, Slack).

Head of Electronics · Node.js & Frontend (React) Dev — SpringLabs / Octopy *Feb 2019 – Dec 2022 · Remote*

Clients: Bimbo (Ricolino & Barcel) · Nestlé · Novo Nordisk · Telefónica · Intel (partner)

- **Head of the Electronics area** (after the rebrand to Octopy): led a team of **2 people**, owning device design, protocols and deliverables.
- **Contributed to the project awarded Intel Partner of the Year #1 in AI** (Intel Partner Connect, May 2020) — press release.
- Designed, built and programmed **autonomous vending robots** with an **embedded management dashboard** for **Bimbo, Nestlé and Novo Nordisk** (firmware + electronics + software).
- **Ricolino** and **Barcel** robots showcased at **Plaza Galerías Toluca** as a Bimbo innovation project.
- Built **internal Node.js chatbots for Telefónica**: meetings management, time tracking and employee check-in/out.

- **React** UIs (including a real-time interface for an autonomous system via Socket.IO); hardware integration with **ROS, C++ and Python**, PCB (Eagle) and RVIZ/Gazebo simulation.

Earlier Experience — Biomedical Engineering & Electronics *2015 – 2019*

- **Junior Researcher — CIR "Dr. Hideyo Noguchi"** (2017–2019): CAD prototyping, biomedical instrumentation, circuit/PCB design, SMD soldering and additive manufacturing.
- **Service Engineer — Biomedical Engineering** (2016–2017): preventive/predictive/corrective maintenance of medical equipment and hospital certification protocols (Cancún).

HIGHLIGHTED PROJECTS

- **Autonomous vending robots (Bimbo · Ricolino & Barcel)** — Mechatronic design, electronics, firmware and embedded dashboard; linked to the **Intel Partner of the Year #1 in AI (2020)** recognition. ▶ watch demo
- **ORTESISCODO (thesis)** — Active elbow-rehabilitation orthosis via additive manufacturing: **embedded firmware** (sensors + actuators) + **Bluetooth configuration app in LabVIEW**.
- **Agenda Inteligente** — Full-stack app with **React 19 + Vite + TailwindCSS** and **Express + MongoDB**, with **OpenAI, Google Gemini and OCR (Tesseract.js)**. GitHub

EDUCATION

B.Sc. in Biomedical Engineering — Universidad Modelo *2012 – 2016*

- Thesis: active elbow-rehabilitation orthosis manufactured via additive manufacturing (firmware + software + prototype).

CERTIFICATIONS

Python · TypeScript · Advanced TypeScript — Platzi · 2022

Database Fundamentals · Node.js · Backend REST API (Express) · PostgreSQL — Platzi · 2022

Salesforce Trailblazer — Salesforce · 2023

LANGUAGES & SOFT SKILLS

Languages:	Spanish (Native) · English (B2) · French (Basic/A2)
Soft skills:	Leadership (team of 2) · Communication · Self-management · Self-learning · Resilience · Teamwork